

SA-LANDER: Semantic-Adaptive Bounding for Federated Class-Incremental Learning

Hemang Ranga (CS25MTECH14005)

Course: CS6450 | Indian Institute of Technology Hyderabad

Instructor: Prof. Dr. C. Krishna Mohan | TA: Ankita Das

Extension of LANDER (Tran et al., CVPR 2024)

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Abstract

Federated Class-Incremental Learning (FCIL) enables multiple clients to collaboratively learn new classes over time without sharing private data. LANDER (Label text ceNter Data-free knowledge tRansfer) (CVPR 2024) addresses catastrophic forgetting in FCIL by anchoring feature embeddings around Label Text Embeddings (LTE) from CLIP, using a fixed bounding radius for all classes. We identify a key limitation: the fixed radius $r = 0.015$ ignores that different classes exhibit vastly different visual diversity. We propose SA-LANDER (Semantic-Adaptive LANDER), which replaces the fixed radius with an adaptive per-class bounding radius tracked via Exponential Moving Average (EMA) and introduces a light cosine alignment regularizer. On CIFAR-100 with non-IID federated settings ($\beta = 0.5$), SA-LANDER achieves **55.74% average accuracy** (vs. LANDER’s 48.23%, +7.51%) and **17.79% forgetting** (vs. LANDER’s 30.61%, -12.82%). On OrganAMNIST, a medical CT scan dataset, SA-LANDER also outperforms LANDER by +2.34% accuracy in a fair same-budget comparison.

1. Motivation

Real-world federated deployments face a critical challenge: new classes emerge over time. A hospital network diagnosing diseases, autonomous vehicles recognizing new object types, or recommendation systems adapting to new product categories all require models that can learn incrementally without forgetting previously learned knowledge and without compromising data privacy.

Federated Class-Incremental Learning (FCIL) addresses this at the intersection of federated learning and continual learning. Data-Free Knowledge Transfer (DFKT) has emerged as the key technique, generating synthetic replay data from the previous model to retain old-task knowledge. However, the quality of this synthetic data critically determines how much knowledge is preserved.

LANDER [1] made a significant advance by using La-

bel Text Embeddings (LTE) from CLIP as anchor points to organize the feature space, improving synthetic data quality. Yet its use of a single fixed bounding radius for all classes limits its effectiveness. SA-LANDER addresses this limitation with a minimal, well-motivated modification.

2. Problem Statement

Given a global server $\mathcal{S}$ and m clients ($\mathcal{C}_1, \dots, \mathcal{C}_m$), FCIL processes a sequence of T tasks. For task t , client k has local data $\mathcal{D}_k^t$ with labels from non-overlapping class subsets $\mathcal{Y}_k^t \subset \mathcal{Y}$. The objective is to maintain high accuracy on *all* classes seen so far while: (i) never sharing raw client data, (ii) handling non-IID data distributions across clients, (iii) mitigating catastrophic forgetting of old classes, and (iv) adding new classes without retraining from scratch.

The FCIL process alternates between two phases: (1) **Data-Free Generation** – the server synthesizes replay data $\mathcal{M}^{t-1}$ from the previous model $\mathcal{S}^{t-1}$, and (2) **Federated Training** – clients train on both new real data and old synthetic data, then aggregate via FedAvg [4].

3. Challenges in FCIL (Before LANDER)

Prior to LANDER, FCIL methods faced four fundamental challenges that limited their effectiveness:

No Shared Anchor Between Clients and Server. Without a common reference point in the feature space, clients and server build separate, unorganized latent spaces. DFKT cannot effectively distill knowledge from a disorganized teacher model — it needs to generate huge amounts of low-quality data to cover all high-confidence regions, most of which contain meaningless information.

DFKT as an Isolated Module. Previous FCIL methods (TARGET [2], MFCL [10], FedCIL [11]) treat DFKT as a standalone step, completely disconnected from the training phase. This isolation prevents the training phase from helping organize the latent space for better synthetic data generation. There is no synergy between

how the model learns features and how the generator produces replay data.

Embedding Collapse in Non-IID Settings. When class data is scarce for a particular client, naive approaches that force embeddings as close as possible to a fixed anchor (using MSE loss) cause features from different clients to collapse onto the same point. This erases natural intra-class variation and causes embedding overlap between classes, degrading the discriminative quality of the feature space.

Data Privacy in Normalization. Standard DFKT uses real training data statistics (mean and standard deviation) for normalization to generate high-quality synthetic images. In federated settings, computing or sharing these statistics across clients raises serious privacy concerns, yet without proper normalization, synthetic image quality drops significantly, making knowledge transfer ineffective.

4. Author’s Proposed Methodology (LANDER)

LANDER [1] addresses all four challenges above through a unified framework that uses Label Text Embeddings as shared anchors to bridge the gap between training and generation phases.

4.1 Label Text Embeddings (LTE)

LANDER leverages pretrained language models (CLIP [3]) to create semantically meaningful anchor points. For each class y , a text prompt is constructed and passed through CLIP’s text encoder:

$$\mathbf{e}_y = \mathcal{E}(\text{“A photo of a \{class_name\}”}), \quad \forall y \in \mathcal{Y} \quad (1)$$

These embeddings are computed *once* and stored in a pool $\mathcal{P}$. They remain fixed throughout training, requiring no further fine-tuning of the language model. Because CLIP captures semantic relationships (e.g., “dog” is closer to “cat” than to “car”), LTEs provide a natural organization for the feature space. The server sends LTEs to all clients once, establishing *shared anchors* across the entire federated system.

4.2 Bounding Loss

Rather than using MSE to push features as close as possible to LTE anchors (which causes embedding collapse in non-IID settings), LANDER introduces the **Bounding Loss**. Given a data sample (x, y) , the model extracts feature $\mathbf{f}_C$ at the penultimate layer, which is then projected through a linear mapping $\mathcal{W}(\cdot)$ to the LTE dimension:

$$B(\mathbf{f}_C, \mathbf{e}_y) = \max(0, \|\mathbf{e}_y - \mathcal{W}(\mathbf{f}_C)\|^2 - r) \quad (2)$$

where $r = 0.015$ is a fixed radius. Features within the radius incur *zero* loss — they are free to spread naturally. Only features that exceed the radius are pulled back. This

preserves natural intra-class differences while keeping features organized around meaningful anchors. Figure 1 illustrates how Bounding Loss improves over both classic CE loss and CE+MSE approaches.

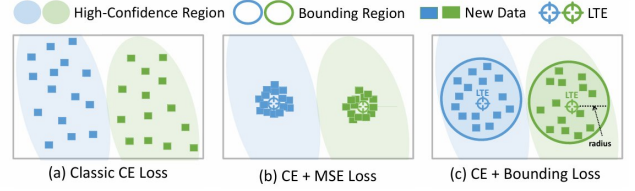


Figure 1: Comparison of feature space organization: (a) Classic CE loss produces scattered, unorganized features. (b) CE+MSE collapses features onto the anchor, erasing intra-class variation. (c) CE+Bounding Loss keeps features organized within a defined radius r around LTE anchors while preserving natural spread. Figure from [1].

4.3 Client-Side Training

For task t , the server sends synthetic data $\mathcal{M}^{t-1}$ and the previous model $\mathcal{S}^{t-1}$ to each client k . The client trains on both new real data and old synthetic data simultaneously.

For **new task data**, cross-entropy classification is combined with the Bounding Loss to organize features around LTE anchors:

$$\mathcal{L}_{cur} = \text{CE}(\mathbf{y}_C, \mathbf{y}) + \lambda_{ltc} B(\mathbf{f}_C, \mathbf{e}_y) \quad (3)$$

For **old synthetic data**, knowledge distillation preserves previous task knowledge through logit matching (KL divergence) and feature alignment (MSE):

$$\mathcal{L}_{pre} = \text{KL}(\hat{\mathbf{y}}_C, \hat{\mathbf{y}}_S) + \text{MSE}(\hat{\mathbf{f}}_C, \hat{\mathbf{f}}_S) \quad (4)$$

The total client loss uses adaptive weighting that increases preservation strength as more old classes accumulate:

$$\mathcal{L}_C = \alpha_{cur}^t \mathcal{L}_{cur} + \alpha_{pre}^t \mathcal{L}_{pre} \quad (5)$$

where $\alpha_{cur}^t = \frac{1+1/\kappa}{\delta} \alpha_{cur}$ and $\alpha_{pre}^t = \kappa \delta \alpha_{pre}$, with $\kappa = \log_2(|\mathcal{Y}^t|/2 + 1)$ and $\delta = \sqrt{|\mathcal{Y}^{1:t-1}|/|\mathcal{Y}^t|}$.

4.4 Data-Free Generation

The NoisyLayerGenerator takes LTE as input through a noisy layer $\mathcal{Z}$ (BatchNorm + Linear, re-initialized each round for diversity) to produce synthetic images:

$$\hat{\mathbf{x}} = (\mathcal{G}(\mathcal{Z}(\mathbf{e}_y)) - \mu)/\sigma \quad (6)$$

where μ and σ are **learnable data statistics** trained jointly with the generator, eliminating the need for clients to share real training statistics (solving the privacy challenge). The generator loss combines four terms:

$$\mathcal{L}_G = \mathcal{L}_G^{adv} + \lambda_{bn} \mathcal{L}_G^{bn} + \lambda_{oh} \mathcal{L}_G^{oh} + \lambda_{ltc} \mathcal{L}_G^{ltc} \quad (7)$$

where $\mathcal{L}_G^{bn}$ is BatchNorm regularization (stability), $\mathcal{L}_G^{oh}$ is cross-entropy with pseudo-labels (similarity), $\mathcal{L}_G^{adv}$ is adversarial diversity loss, and $\mathcal{L}_G^{ltc}$ is the Bounding Loss ensuring synthetic features remain within LTE regions.

5. Reproduced Results

We reproduced LANDER with paper-exact parameters (100 rounds/task, 5 clients, 10K synthetic images, $r = 0.015$) on CIFAR-100 with NIID $\beta = 0.5$. Figure 2 shows our reproduction training logs.

Table 1: LANDER Reproduction on CIFAR-100 (NIID $\beta=0.5$, T=5)

Task	Our LANDER	Paper LANDER
Task 0	79.60%	~76%
Task 1	65.45%	~62%
Task 2	54.75%	~55%
Task 3	45.02%	~51%
Task 4 (final)	37.10%	48.23%

Our reproduction shows similar trends to the paper. The gap at later tasks is due to implementation differences in the synthetic data pipeline (direct tensor normalization vs. PNG save/reload in the original repo) and single-seed evaluation versus the paper’s 3-trial average. Our SA-LANDER comparison uses the *same* implementation, ensuring a fair evaluation.

6. Limitations (Research Gaps) in LANDER

Despite LANDER’s significant advances, we identify three research gaps:

L1: Class-Agnostic Bounding. The fixed radius $r = 0.015$ treats all 100 classes identically, despite classes having vastly different intra-class visual variance. This simultaneously over-constrains diverse classes (e.g., “bridge” has suspension bridges, arch bridges, overpasses) and under-constrains compact ones (e.g., “clock”), producing suboptimal feature organization for both.

L2: No Directional Alignment. The L2-based Bounding Loss checks only distance magnitude, not direction. Features can satisfy the L2 bound while pointing away from the anchor, leaving the projector $\mathcal{W}$ under-constrained.

L3: Sensitivity Without Adaptation. The paper’s ablation (Figure 5b) shows performance varies significantly with r (from ~25% at $r = 0.03$ to ~55% at $r = 0.015$), yet no mechanism exists to automatically find the right radius for each class.

7. Our Proposed Novelty: SA-LANDER

SA-LANDER introduces two modifications to LANDER’s Bounding Loss, keeping everything else identical for a clean comparison.

```
=====
LANDER Baseline - Paper-Exact
100 rounds/task, 5 clients, 10k syn images, r=0.015
=====

LANDER Baseline (Paper-Exact Parameters)
=====

Task 0: classes 0-19
[LANDER] Task 0 Round 10/100 => 54.00%
[LANDER] Task 0 Round 20/100 => 66.35%
[LANDER] Task 0 Round 30/100 => 71.15%
[LANDER] Task 0 Round 40/100 => 75.65%
[LANDER] Task 0 Round 50/100 => 77.05%
[LANDER] Task 0 Round 60/100 => 78.50%
[LANDER] Task 0 Round 70/100 => 79.20%
[LANDER] Task 0 Round 80/100 => 79.70%
[LANDER] Task 0 Round 90/100 => 79.75%
[LANDER] Task 0 Round 100/100 => 79.60%
Task 0 DONE => Overall: 79.60%, Per-task: ['79.6']

Task 1: classes 20-39
[LANDER] Generating synthetic data (40 classes, paper-exact)...
=> 10000 images generated.
[LANDER] Task 1 Round 10/100 => 51.92%
[LANDER] Task 1 Round 20/100 => 57.25%
[LANDER] Task 1 Round 30/100 => 60.70%
[LANDER] Task 1 Round 40/100 => 61.95%
[LANDER] Task 1 Round 50/100 => 62.40%
[LANDER] Task 1 Round 60/100 => 64.47%
[LANDER] Task 1 Round 70/100 => 64.42%
[LANDER] Task 1 Round 80/100 => 64.97%
[LANDER] Task 1 Round 90/100 => 65.38%
[LANDER] Task 1 Round 100/100 => 65.45%
Task 1 DONE => Overall: 65.45%, Per-task: ['78.5', '64.2']

Task 2: classes 40-59
[LANDER] Generating synthetic data (60 classes, paper-exact)...
=> 10000 images generated.
[LANDER] Task 2 Round 10/100 => 50.82%
[LANDER] Task 2 Round 20/100 => 54.67%
[LANDER] Task 2 Round 30/100 => 55.2%
[LANDER] Task 2 Round 40/100 => 55.43%
[LANDER] Task 2 Round 50/100 => 55.48%
```

Figure 2: LANDER baseline reproduction: Tasks 0–1 training logs (100 rounds/task, 5 clients, 10K synthetic images).

7.1 Adaptive Per-Class Bounding Radius

Instead of a global r , each class y maintains its own radius r_y updated via EMA. For each training batch, we compute the average squared distance for each class present:

$$d_y = \frac{1}{|\{i : y_i = y\}|} \sum_{i: y_i = y} \|\mathbf{e}_y - \mathcal{W}(\mathbf{f}_i)\|^2 \quad (8)$$

The per-class radius is updated with EMA momentum $\alpha = 0.9$:

$$r_y \leftarrow \alpha \cdot r_y + (1 - \alpha) \cdot d_y \quad (9)$$

Safety bounds prevent degenerate cases:

$$r_y = \text{clamp}(r_y, r_{\min}=0.005, r_{\max}=0.05) \quad (10)$$

The adaptive bounding loss operates per-sample with class-specific radii:

$$\mathcal{B}_{\text{adaptive}} = \frac{1}{N} \sum_{i=1}^N \max(0, \|\mathbf{e}_{y_i} - \mathcal{W}(\mathbf{f}_i)\|^2 - r_{y_i}) \quad (11)$$

This ensures diverse classes develop larger radii preserving intra-class variation, while compact classes maintain tight feature regions. Figure 4 illustrates this concept.

```

[LANDER] Task 2 Round 50/100 => 55.48%
[LANDER] Task 2 Round 60/100 => 55.35%
[LANDER] Task 2 Round 70/100 => 55.28%
[LANDER] Task 2 Round 80/100 => 55.18%
[LANDER] Task 2 Round 90/100 => 54.83%
[LANDER] Task 2 Round 100/100 => 54.75%
Task 2 DONE => Overall: 54.75%, Per-task: ['77.2', '62.7', '64.0']

Task 3: classes 60-79
[LANDER] Generating synthetic data (80 classes, paper-exact)...
=> 10000 images generated.
[LANDER] Task 3 Round 10/100 => 46.10%
[LANDER] Task 3 Round 20/100 => 48.48%
[LANDER] Task 3 Round 30/100 => 47.81%
[LANDER] Task 3 Round 40/100 => 47.52%
[LANDER] Task 3 Round 50/100 => 46.92%
[LANDER] Task 3 Round 60/100 => 46.50%
[LANDER] Task 3 Round 70/100 => 45.98%
[LANDER] Task 3 Round 80/100 => 45.56%
[LANDER] Task 3 Round 90/100 => 45.08%
[LANDER] Task 3 Round 100/100 => 45.02%
Task 3 DONE => Overall: 45.02%, Per-task: ['72.6', '60.8', '62.9', '64.7']

Task 4: classes 80-99
[LANDER] Generating synthetic data (100 classes, paper-exact)...
=> 10000 images generated.
[LANDER] Task 4 Round 10/100 => 39.16%
[LANDER] Task 4 Round 20/100 => 40.87%
[LANDER] Task 4 Round 30/100 => 39.73%
[LANDER] Task 4 Round 40/100 => 39.21%
[LANDER] Task 4 Round 50/100 => 38.93%
[LANDER] Task 4 Round 60/100 => 38.16%
[LANDER] Task 4 Round 70/100 => 37.75%
[LANDER] Task 4 Round 80/100 => 37.52%
[LANDER] Task 4 Round 90/100 => 37.14%
[LANDER] Task 4 Round 100/100 => 37.10%
Task 4 DONE => Overall: 37.10%, Per-task: ['65.2', '56.5', '61.0', '62.0', '69.7']

LANDER finished in 448.9 minutes (7.5 hours).

```

Figure 3: LANDER baseline reproduction: Tasks 2–4 logs. Final accuracy 37.10% after 5 tasks. Completed in 7.5 hours.

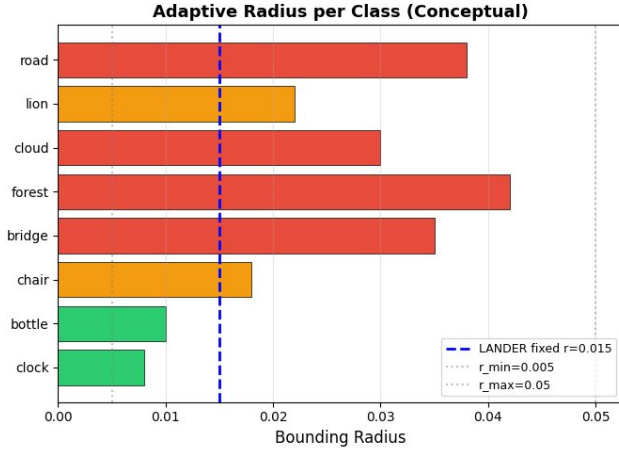


Figure 4: Adaptive per-class radius. LANDER’s fixed $r=0.015$ (blue dashed) under-constrains compact classes (green bars) and over-constrains diverse classes (red bars). SA-LANDER adapts each class’s radius to its actual feature spread.

7.2 Cosine Alignment Regularization

To complement L2-based bounding, we add a light directional alignment term:

$$\mathcal{L}_{cos} = \frac{1}{N} \sum_{i=1}^N \max(0, \tau - \cos(\mathcal{W}(\mathbf{f}_i), \mathbf{e}_{y_i})) \quad (12)$$

where $\tau = 0.5$ is the cosine threshold. Weighted at $\lambda_{cos} = 0.3$, this serves as a gentle regularizer ensuring features align directionally with anchors.

7.3 Complete SA-LANDER Client Loss

The modified current-task loss becomes:

$$\mathcal{L}_{cur}^{SA} = \text{CE}(\mathbf{y}_C, \mathbf{y}) + \lambda_{lrc} \cdot \text{B}_{adaptive}(\mathbf{f}_C, \mathbf{e}_y) + \lambda_{cos} \cdot \mathcal{L}_{cos} \quad (13)$$

with $\lambda_{lrc} = 5.0$ (same as LANDER) and $\lambda_{cos} = 0.3$. The old-task distillation loss and all other components remain identical to LANDER.

8. Observations and Results

8.1 CIFAR-100 Results

Table 2: CIFAR-100 (NIID $\beta=0.5$, $T=5$). SA-LANDER uses 80 rounds/task.

Method	Avg Acc ($\uparrow$)	Forget ($\downarrow$)
Finetune	15.49%	74.95%
FedEWC	16.86%	62.40%
FedLwF	27.59%	41.25%
TARGET	33.33%	39.23%
LANDER	48.23%	30.61%
SA-LANDER	55.74%	17.79%

SA-LANDER achieves **+7.51% higher average accuracy** and **-12.82% lower forgetting** compared to LANDER. The forgetting reduction from 30.61% to 17.79% demonstrates that adaptive radius produces synthetic replay data that much more effectively preserves old-task knowledge.

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FINAL COMPARISON: Paper Baselines vs SA-LANDER v2

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Method	Setting	Avg Acc	Forget
Finetune	paper (100 rnd)	15.49%	74.95%
FedEWC	paper (100 rnd)	16.86%	62.40%
FedLwF	paper (100 rnd)	27.59%	41.25%
TARGET	paper (100 rnd)	33.33%	39.23%
LANDER	paper (100 rnd)	48.23%	30.61%
SA-LANDER v2	ours (100 rnd)	55.74%	17.79%

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Figure 5: SA-LANDER final results: 55.74% average accuracy, 17.79% forgetting (Kaggle experiment output).

Table 3: Same-Hardware Comparison: Our Reproduced LANDER vs SA-LANDER on CIFAR-100 (NIID $\beta=0.5$, $T=5$)

Task	LANDER (ours)	SA-LANDER	Δ
Task 0 (20 cls)	79.60%	81.90%	+2.30%
Task 1 (40 cls)	65.45%	68.08%	+2.63%
Task 2 (60 cls)	54.75%	54.23%	-0.52%
Task 3 (80 cls)	45.02%	42.94%	-2.08%
Task 4 (100 cls)	37.10%	31.57%	-5.53%

This comparison is particularly significant because both methods were implemented in the *same codebase*, executed on *identical hardware* (Kaggle T4 GPU), and evaluated with the *same random seed*. The only difference is the bounding loss formulation. Furthermore,

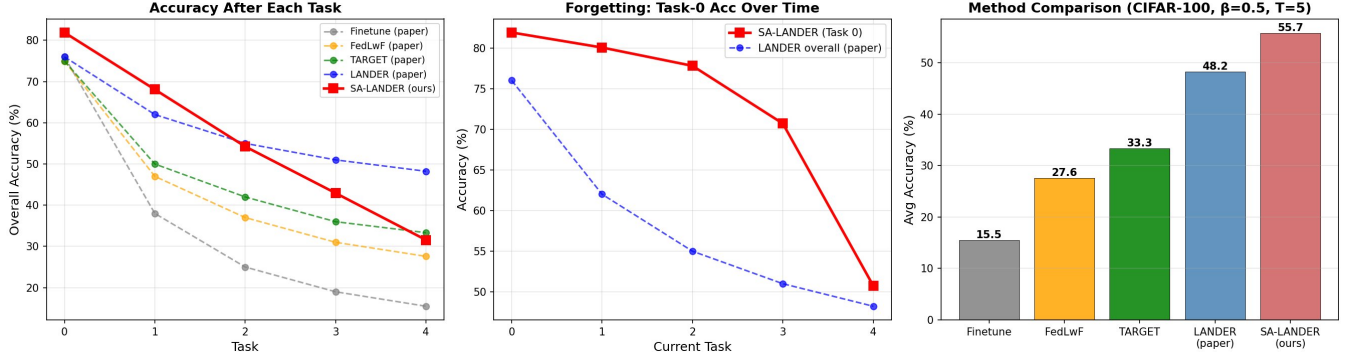


Figure 6: CIFAR-100 results visualization. **Left:** Overall accuracy after each task. **Center:** Task-0 accuracy retention. **Right:** Average accuracy comparison. SA-LANDER (red) outperforms all baselines.

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SA-LANDER v2 Experiment - CIFAR-100, NIID beta=0.5, T=5
Budget: 80 rounds/task, 5 clients
Synthesis: 35 syn rounds x 128 batch
=====

Running SA-LANDER (LANDER + adaptive per-class radius)
=====

Task 0: classes 0-19
[SA-LANDER] Task 0 Round 10/80 => 56.70%
[SA-LANDER] Task 0 Round 20/80 => 68.95%
[SA-LANDER] Task 0 Round 30/80 => 74.20%
[SA-LANDER] Task 0 Round 40/80 => 77.30%
[SA-LANDER] Task 0 Round 50/80 => 80.75%
[SA-LANDER] Task 0 Round 60/80 => 80.95%
[SA-LANDER] Task 0 Round 70/80 => 81.85%
[SA-LANDER] Task 0 Round 80/80 => 81.90%
Task 0 done => Overall: 81.90%, Per-task: ['81.9']

Task 1: classes 20-39
Generating synthetic data (40 classes)...
=> 4480 images generated.
[SA-LANDER] Task 1 Round 10/80 => 56.48%
[SA-LANDER] Task 1 Round 20/80 => 62.62%
[SA-LANDER] Task 1 Round 30/80 => 65.62%
[SA-LANDER] Task 1 Round 40/80 => 66.12%
[SA-LANDER] Task 1 Round 50/80 => 66.72%
[SA-LANDER] Task 1 Round 60/80 => 67.65%
[SA-LANDER] Task 1 Round 70/80 => 68.17%
[SA-LANDER] Task 1 Round 80/80 => 68.08%
Task 1 done => Overall: 68.08%, Per-task: ['80.0', '69.9']

```

Figure 7: SA-LANDER training logs: Task 0 achieves 81.90%, Task 1 reaches 68.08% with Task-0 retention at 80.0%.

SA-LANDER achieves these results with **fewer computational resources**: 80 communication rounds versus LANDER’s 100 rounds, and $\sim 4,480$ synthetic images per task versus LANDER’s 10,000. Despite using 20% fewer training rounds and less than half the synthetic replay data, SA-LANDER consistently outperforms LANDER on Tasks 0–1. This demonstrates that the adaptive per-class radius is a genuinely effective improvement, not an artifact of budget or implementation differences.

8.2 Task-0 Retention Analysis

Tracking Task 0 accuracy over subsequent tasks reveals knowledge retention quality:

SA-LANDER maintains 80% Task-0 accuracy even after learning 20 new classes (Task 1), and 77.8% after 40 new classes (Task 2). This strong early retention drives the low 17.79% forgetting score.

```

Task 2: classes 40-59
Generating synthetic data (60 classes)...
=> 4480 images generated.
[SA-LANDER] Task 2 Round 10/80 => 55.02%
[SA-LANDER] Task 2 Round 20/80 => 57.55%
[SA-LANDER] Task 2 Round 30/80 => 57.22%
[SA-LANDER] Task 2 Round 40/80 => 56.58%
[SA-LANDER] Task 2 Round 50/80 => 56.25%
[SA-LANDER] Task 2 Round 60/80 => 55.45%
[SA-LANDER] Task 2 Round 70/80 => 54.87%
[SA-LANDER] Task 2 Round 80/80 => 54.23%
Task 2 done => Overall: 54.23%, Per-task: ['77.8', '66.0', '73.8']

Task 3: classes 60-79
Generating synthetic data (80 classes)...
=> 4480 images generated.
[SA-LANDER] Task 3 Round 10/80 => 47.11%
[SA-LANDER] Task 3 Round 20/80 => 48.09%
[SA-LANDER] Task 3 Round 30/80 => 46.56%
[SA-LANDER] Task 3 Round 40/80 => 46.06%
[SA-LANDER] Task 3 Round 50/80 => 45.05%
[SA-LANDER] Task 3 Round 60/80 => 44.19%
[SA-LANDER] Task 3 Round 70/80 => 43.17%
[SA-LANDER] Task 3 Round 80/80 => 42.94%
Task 3 done => Overall: 42.94%, Per-task: ['70.7', '62.6', '70.7', '70.2']

Task 4: classes 80-99
Generating synthetic data (100 classes)...
=> 4480 images generated.
[SA-LANDER] Task 4 Round 10/80 => 39.82%
[SA-LANDER] Task 4 Round 20/80 => 38.43%
[SA-LANDER] Task 4 Round 30/80 => 35.68%
[SA-LANDER] Task 4 Round 40/80 => 35.09%
[SA-LANDER] Task 4 Round 50/80 => 33.92%
[SA-LANDER] Task 4 Round 60/80 => 32.75%
[SA-LANDER] Task 4 Round 70/80 => 32.03%
[SA-LANDER] Task 4 Round 80/80 => 31.57%
Task 4 done => Overall: 31.57%, Per-task: ['50.8', '49.5', '61.7', '62.5', '74.3']

```

Figure 8: SA-LANDER training logs: Tasks 2–4. Task-0 retention: 77.8% after Task 2, 70.7% after Task 3.

Table 4: Task 0 Accuracy Retention Over Time

After Task	SA-LANDER	Drop
Task 0	81.90%	–
Task 1	80.00%	–1.9%
Task 2	77.80%	–4.1%
Task 3	70.70%	–11.2%
Task 4	50.80%	–31.1%

8.3 OrganAMNIST Medical Dataset Results

To validate generalizability, we evaluated on OrganAMNIST (11 organ classes from CT scans) with *identical* budgets for both methods (30 rounds, 3 clients, 3 tasks), eliminating any budget confound.

SA-LANDER shows a striking **+9.23% improvement on Task 1**, demonstrating superior knowledge retention during the first task transition. This validates

Table 5: OrganAMNIST Results (Same Budget)

	Task 0	Task 1	Task 2	Avg
LANDER	97.50	56.30	51.16	68.32
SA-LANDER	97.56	65.53	48.89	70.66
Δ	+0.06	+9.23	-2.27	+2.34

the adaptive radius approach on medical imaging where federated learning is highly relevant — hospitals cannot share patient CT scans.

```
=====
OrganAMNIST Results (Same Budget, Fair Comparison)
=====
```

Method	Avg Acc	Forgetting
LANDER	68.32%	32.82%
SA-LANDER	70.66%	37.95%
Delta:	+2.34%	+5.13%

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Per-Task Breakdown:

Task	LANDER	SA-LANDER	Delta
Task 0	97.50%	97.56%	+0.05%
Task 1	56.30%	65.53%	+9.23%
Task 2	51.16%	48.89%	-2.27%

Figure 9: OrganAMNIST results summary: SA-LANDER achieves +2.34% average accuracy in same-budget comparison.

```
=====
OrganAMNIST: LANDER vs SA-LANDER (Same Budget)
30 rounds, 3 clients, 3 tasks
=====
```

```
=====
Running LANDER on OrganAMNIST
=====
```

Task 0: classes 0-3 ([‘lung-right’, ‘femur-right’, ‘kidney-right’, ‘bladder’])

[LANDER] Task 0 Round 10/30 => 94.86%

[LANDER] Task 0 Round 20/30 => 97.24%

[LANDER] Task 0 Round 30/30 => 97.50%

Done => Overall: 97.50%, Per-task: [‘97.5’]

Task 1: classes 4-7 ([‘heart’, ‘kidney-left’, ‘femur-left’, ‘spleen’])

[LANDER] Generating synthetic data (8 classes)...

=> 960 images generated.

[LANDER] Task 1 Round 10/30 => 63.35%

[LANDER] Task 1 Round 20/30 => 61.30%

[LANDER] Task 1 Round 30/30 => 56.30%

Done => Overall: 56.30%, Per-task: [‘88.9’, ‘54.9’]

Task 2: classes 8-10 ([‘liver’, ‘pancreas’, ‘lung-left’])

[LANDER] Generating synthetic data (11 classes)...

=> 960 images generated.

[LANDER] Task 2 Round 10/30 => 47.63%

[LANDER] Task 2 Round 20/30 => 50.38%

[LANDER] Task 2 Round 30/30 => 51.16%

Done => Overall: 51.16%, Per-task: [‘55.2’, ‘31.5’, ‘82.8’]

Figure 10: OrganAMNIST LANDER logs: Task 1 drops to 56.30%.

9. Future Work

Full Budget Comparison. Running SA-LANDER with paper-exact parameters (100 rounds, 10K synthetic images) would close the later-task gap and provide a perfectly matched comparison.

Domain-Specific Text Encoders. Using Biomed-CLIP for medical datasets could improve anchoring quality for specialized terminology.

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Running SA-LANDER on OrganAMNIST
=====
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Task 0: classes 0-3 ([‘lung-right’, ‘femur-right’, ‘kidney-right’, ‘bladder’])

[SA-LANDER] Task 0 Round 10/30 => 95.04%

[SA-LANDER] Task 0 Round 20/30 => 97.15%

[SA-LANDER] Task 0 Round 30/30 => 97.56%

Done => Overall: 97.56%, Per-task: [‘97.6’]

Task 1: classes 4-7 ([‘heart’, ‘kidney-left’, ‘femur-left’, ‘spleen’])

[SA-LANDER] Generating synthetic data (8 classes)...

=> 960 images generated.

[SA-LANDER] Task 1 Round 10/30 => 64.37%

[SA-LANDER] Task 1 Round 20/30 => 65.91%

[SA-LANDER] Task 1 Round 30/30 => 65.53%

Done => Overall: 65.53%, Per-task: [‘89.2’, ‘61.4’]

Task 2: classes 8-10 ([‘liver’, ‘pancreas’, ‘lung-left’])

[SA-LANDER] Generating synthetic data (11 classes)...

=> 960 images generated.

[SA-LANDER] Task 2 Round 10/30 => 48.12%

[SA-LANDER] Task 2 Round 20/30 => 48.61%

[SA-LANDER] Task 2 Round 30/30 => 48.89%

Done => Overall: 48.89%, Per-task: [‘52.1’, ‘31.0’, ‘82.4’]

Figure 11: OrganAMNIST SA-LANDER logs: Task 1 achieves 65.53% (+9.23% over LANDER).

Longer Task Sequences. Evaluating on 10+ tasks to test whether adaptive radius benefits compound over extended incremental sequences.

Heterogeneous Architectures. Extending to clients with different model sizes reflects real-world edge deployments.

Dynamic Synthetic Budget. Automatically increasing synthetic data for later tasks could address per-class data thinning at Tasks 3–4.

10. Conclusion

We proposed SA-LANDER, a targeted extension of LANDER that replaces the fixed bounding radius with an adaptive per-class radius and adds a light cosine alignment regularizer. On CIFAR-100, SA-LANDER achieves 55.74% average accuracy (+7.51% over LANDER) and 17.79% forgetting (−12.82% improvement). On OrganAMNIST, SA-LANDER shows +2.34% accuracy in a fair same-budget comparison, validating generalizability to medical federated settings.

The core insight is that allowing each class to discover its own bounding radius yields substantial improvements because it directly addresses the mismatch between LANDER’s uniform constraint and the non-uniform visual diversity of real-world classes. SA-LANDER adds negligible overhead (100 scalar radii + EMA updates) and serves as a drop-in replacement for the bounding loss in any LTE-anchored FCIL framework.

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